

MIDTERM REVIEW

Binary Conversions, Combinational Logic Design, Arithmetic
Hardware, Binary Arithmetic

Notes

- This exam requires the Respondus LockDown Browser. Your Camera on.
- 9 questions
- No open notes.
- Materials to use
 - Pen/Pencil
 - A non-graphing calculator
 - Scratch paper

Sample Question.1

- Let $A = (1100\ 1111)_2$. What decimal value does this represent if A is unsigned?
- Solution:
- $A = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b^1 + a_0 b^0$
- $A = 1*b^7 + 1*b^6 + 0*b^5 + 0*b^4 + 1*b^3 + 1*b^2 + 1*b + 1$
- $A = 128 + 64 + 0 + 0 + 8 + 4 + 2 + 1$
- $A = 207$

Sample Question.2

- Convert $(55)_{10}$ to binary (unsigned). Show your answer using 8 bits.
- Solution:
- $55 = 2 \cdot 27 + 1$ ← the remainder here is the least significant bit.
- $27 = 2 \cdot 13 + 1$
- $13 = 2 \cdot 6 + 1$
- $6 = 2 \cdot 3 + 0$
- $3 = 2 \cdot 1 + 1$
- $1 = 2 \cdot 0 + 1$ ← continue until the quotient becomes 0.
- $(55)_{10} = (0011\ 0111)_2$

Sample Question.3

- Let $A = (0100\ 1111)_2$. What decimal value does this represent if A is represented with the 2's Complement representation?
- Solution:
- $A = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b^1 + a_0 b^0$
- $A = 1 * b^6 + 0 * b^5 + 0 * b^4 + 1 * b^3 + 1 * b^2 + 1 * b + 1$
- $A = 64 + 0 + 0 + 8 + 4 + 2 + 1$
- $A = 79$

Sample Question.4

- Let $A = (1100\ 1111)_2$. What decimal value does this represent if A is represented with the 2's Complement representation?
- Solution:
- 0011 0001
- $n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b^1 + a_0 b^0$
- $n = 0*b^7 + 0*b^6 + 1*b^5 + 1*b^4 + 0*b^3 + 0*b^2 + 0*b + 1$
- $n = 32 + 16 + 0 + 0 + 0 + 1$
- $n = 49$
- $A = -49$

Sample Question.5

- Convert $(49)_{10}$ to binary using the 2's complement representation of signed integers. Show your answer using 8 bits.
- Solution:
- $49 = 2 * 24 + 1$
- $24 = 2 * 12 + 0$
- $12 = 2 * 6 + 0$
- $6 = 2 * 3 + 0$
- $3 = 2 * 1 + 1$
- $1 = 2 * 0 + 1$
- $(49)_{10} = (110001)_2 = (0011\ 0001)_2$

Sample Question.6

- Convert $(-55)_{10}$ to binary using the 2's complement representation of signed integers. Show your answer using 8 bits.

- Solution:

- $55 = 2 * 27 + 1$
- $27 = 2 * 13 + 1$
- $13 = 2 * 6 + 1$
- $6 = 2 * 3 + 0$
- $3 = 2 * 1 + 1$
- $1 = 2 * 0 + 1$
- $(55)_{10} = (110111)_2$

$$(55)_{10} = (110111)_2$$

$$(55)_{10} = (0011\ 0111)_2$$

$$(-55)_{10} = (1100\ 1001)_2$$

Sample Question.7

- Suppose $A = 0101$ and $B = 0001$. Calculate $A-B$ and show your answer using 4 bits. State whether or not overflow has occurred.

- Solution:
- $-B \rightarrow 1110$ (bitwise inverse) $+ 1 \rightarrow 1111$

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Result: 0100
Overflow: No

Sample Question.8

- Consider the following Karnaugh map. Which of the following equations represents a most simplified sum-of-products equation for F?

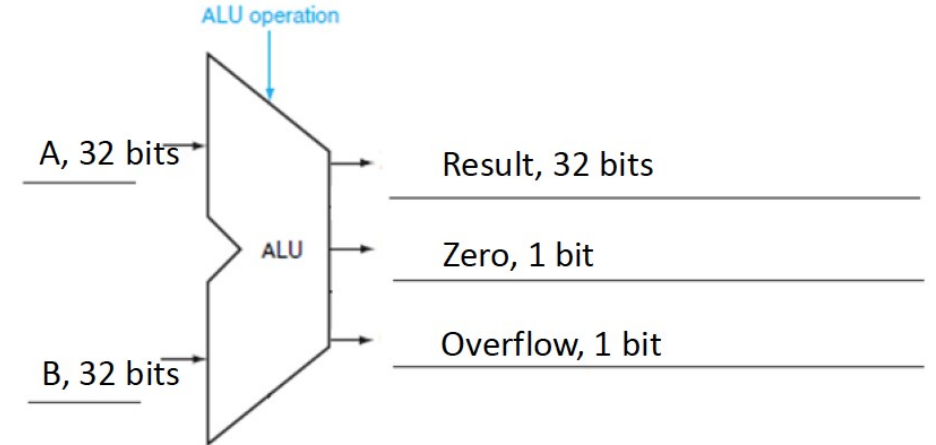
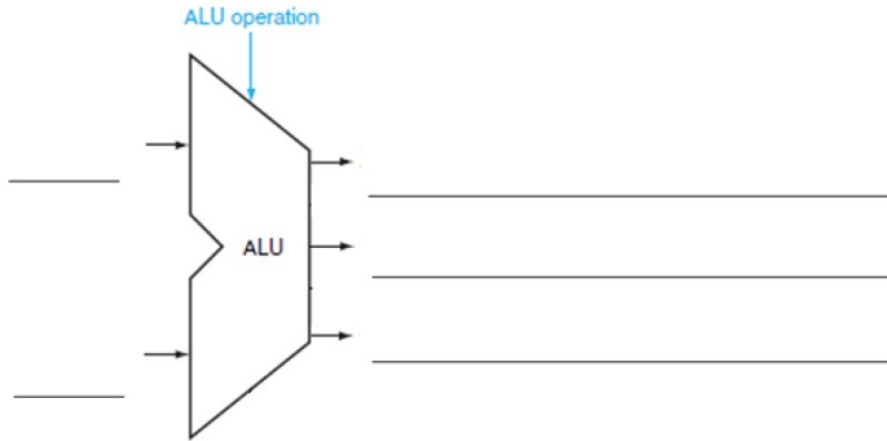
		A	
C	0	0	1
	0	1	1
	1	0	0
	1	0	1
		B	

		A	
C	0	0	1
	0	1	1
	1	0	0
	1	0	1
		B	

$$F = AC' + AB' + A'BC$$

Sample Question.9

- Label the inputs and outputs of the following 32-bit Arithmetic Logic Unit. Identify how many bits would be used to represent each value.



Sample Question.10

- Multiply 2 x -5 using Booth's algorithm. Use 4-bit for multiplicand (2) and multiplier (-5).
- Multiplier: 1011
- Multiplicand: 0010
- -Multiplicand: ?

Solution:

-Multiplicand = 1110

$$(0000\ 1001)_2 + 1 = 10$$

Iteration	Step	Product	Previous LSB	Action
0	-	0000 1011	0	Initialize
1	1.10	1110 1011	0	Subtract
1	1.2	1111 0101	1	Shift
2	2.11	1111 0101	1	No action
2	2.2	1111 1010	1	Shift
3	3.01	0001 1010	1	Add
3	3.2	0000 1101	0	Shift
4	4.11	1110 1101	1	Subtract
4	4.2	1111 0110	0	Shift

Sample Question.11

- Automatic Light with Motion Detection
- Consider the design of an automatic outdoor light for a house. This light will be on if there is not enough light to see by otherwise or if motion is detected near the light. This requires two sensors: a motion sensor and a daylight sensor. The light can also be turned on and off by a light switch.
- The basic operation of the light and the inputs and outputs of the controller are shown below.
- The inputs and outputs are:
 - L = Light (1 = On, 0 = Off)
 - D = Daylight Sensor (1 = do not need light, 0 = dark, light is required)
 - M = Motion Detector (1 = Motion Detected, 0 = No Motion Detected)
 - S = Active switch (1 = light is on manually, 0 = light is off, rely on sensors)
- The basic operation of the alarm can be described as:
 - If active switch is off, check daylight sensor and motion detector.
 - If daylight sensor is "dark" (W=0) or motion detector detects motion (M=1), turn on lights (L=1).

Inputs			Output
S (Active switch)	M (Motion Detector)	D (Daylight Sensor)	L (Lights)
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Sample Question.11 (Cont'd)

Inputs			Output
S (Active switch)	M (Motion Detector)	D (Daylight Sensor)	L (Lights)
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

D \ SM	00		01		11		10	
	S	M	S	M	S	M	S	M
0	1	0	1	0	1	0	1	0
1	0	1	1	1	1	1	1	1

$$L = S + M + D'$$

